

Youngsam Chun

Responsible AI Researcher · AI Innovation, Trustworthy AI, and Moral Value Alignment

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PROFILE

Researcher specialising in the mechanisms of AI innovation, with expertise in trustworthy and responsible artificial intelligence, moral value alignment, and system reliability. Current work bridges responsible AI governance, pluralistic value alignment for generative AI, and risk taxonomies for Physical AI, combining patent analytics, scientometrics, network science, and sentence-embedding-based text analysis.

RESEARCH AREAS

- Responsible AI, trustworthy AI, AI safety, and AI governance
- Generative AI, large language models, RAG systems, and value alignment
- Physical AI risk taxonomy, embodied AI, and cyber-physical system safety
- Agentic RAG security, black-box attacks, and retrieval poisoning
- Patent analytics, scientometrics, and AI innovation measurement
- Economic complexity, technological specialisation, and national competitiveness
- Network science, knowledge-space modelling, and semantic embedding methods

EDUCATION AND TRAINING

Ph.D., Technology Management, Economics, and Policy (TEMEP)

Mar 2019 – Aug 2024

Seoul National University, Seoul, Korea

Final grade 4.28/4.5 (97.8); EQF 8

Thesis: *Exploring complementary innovation of AI technology from a multidimensional knowledge network approach* (HAL: tel-05047941)

M.Sc., Technology Management, Economics, and Policy (TEMEP)

Mar 2017 – Feb 2019

Seoul National University, Seoul, Korea

Final grade 4.33/4.5 (98.3); EQF 7

Thesis: *Heterophily Effects on Industrial Innovation Leadership Changes in Autonomous Vehicle Industry*

Economic Networks Summer School

Jun 2024

University of Oxford, Oxford, United Kingdom — complex network analysis, economic complexity and geography.

Masterclass: MSCA Post-Doctoral Fellowships

May 2025

Université Paris 1 Panthéon-Sorbonne, Paris, France — intensive proposal-writing workshop.

WORK EXPERIENCE

Responsible AI Researcher — Responsible AI Department, Frontier AI Lab, KT Corporation

Aug 2024 – Present

Seoul, Korea

- Establishment of a responsible AI governance framework for high-risk domains.
- Joint research and development of a Korea-specific custom GPT-4o with Microsoft.
- Development of value-alignment technology reflecting Korea's cultural context.

University Lecturer in Business Analytics — UNIST

Aug 2024 – Present

Graduate School of Technology and Innovation Management, Ulsan National Institute of Science and Technology, Ulsan, Korea

- Graduate lectures on business analytics and on deep/machine learning using Python and R.
- Supervision of graduate students in thesis writing and research.

SELECTED PROJECTS

Physical AI Risk Taxonomy

2026 – Present

Project Lead, KT R&D Center (Korea). Interactive L4 risk-card taxonomy for Physical AI systems (embodied AI, humanoids, robots, drones, autonomous vehicles, cyber-physical systems); a structured evidence map of risks arising when AI perceives, decides, and acts in the physical world. Release includes 182 L4 risk cards, bilingual definitions, Korean evidence justifications, reference links, curation data, audit logs, and methodology files. [Live site](#) · [Repository](#)

Contextual Value Alignment through Genre-Based Aesthetic

Judgment

2026 – Present

International collaboration with Isa Weisha Wang (Institute for Ethics in AI, Technical University of Munich). A framework for measuring where large language model evaluators diverge from human interpretive communities in aesthetic judgment, using film genres as measurable cultural contexts. It distinguishes universal from context-dependent value dimensions and locates human and AI gaps through a sign-flip test and a subspace-resolved orientation and calibration audit. Companion to the orientation and calibration work, with a joint manuscript in preparation.

AI Topic Space: Policy, Science, and Technology in the Korean AI

Knowledge Ecosystem

2025 – Present

Principal investigator; joint study with STEPI and KDI (Korea). Interactive semantic map of AI topics across policy reports, academic papers, and patents, built on a four-level topic hierarchy (analytical domain, thematic family, semantic cluster, key phrase). Supports reference-space exploration, Korea activation, and relative topic-gap analysis across five-year periods. [Live site](#) · [Repository](#) · [HuggingFace dataset](#)

Adversarial Attack and Defence Mechanisms for RAG Systems

Oct 2025 – Nov 2026

Led at KT Frontier AI Lab (Responsible AI Department), in collaboration with Korea University. Attack and defence methods for Retrieval-Augmented Generation systems, producing benchmark datasets, an open-source toolkit, and a patent application; bridging academic research with B2B RAG services.

Innovation Characteristics of Generative AI and Large Language

Models: A Patent Analytics Approach

Jul 2025 – Jan 2026

Research Collaborator, KT Frontier AI Lab (Responsible AI Department), in collaboration with STEPI. Sentence-BERT embeddings (PatentSBERTa) and ensemble modelling (XGBoost + MLP) with technological-distance clustering to assess dominant design emergence and cross-national competitive dynamics.

Analysis of Industrial AI Innovation and National

Competitiveness: A Scientometric Approach

Dec 2025 – Jan 2026

Research Collaborator, Seoul National University & Korea Development Institute (KDI). Patent analytics and the Economic Complexity Index (RCA-based) to study domain specialisation and provide policy frameworks for trustworthy AI governance and industrial co-evolution.

PUBLICATIONS AND PREPRINTS

KidnapRAG: A Black-Box Attack for Hijacking Reasoning in

Agentic Retrieval-Augmented Generation Systems

2026

Choi, C., Kim, E., Lee, K., **Chun, Y.**, Jeong, J., Kim, E., Oh, M., Jang, J., & Chang, B. (2026). arXiv preprint. Industry-academia collaboration with Prof. Buru Chang's group (Korea University, lead author); **Chun, Y.** is the KT-side first author. ACL ARR 2026 May submission. [arXiv:2607.00422](#) · [Code](#)

Silent Failure of Moral Alignment in Large Language-Model

Evaluators

2026

Chun, Y. *Under review at Scientific Reports* (June 2026). Reproducibility release: [deep1003/silent-failure-moral-alignment](#).

Moral Orientation and Calibration: Coupled in Human

Annotators, Separable in Judge LLMs

2026

Chun, Y. (2026). Moral Orientation and Calibration: Coupled in Human Annotators, Separable in Judge LLMs. *ICML 2026 Pluralistic Alignment Workshop* (accepted). [Paper](#) · [Code](#)

When Fear Becomes Beauty: Genre as Context in Human and AI Aesthetic Judgment 2026

Wang, I. W., & **Chun, Y.** (2026). Manuscript in preparation. A technical framework for contextual value alignment of large language model evaluators in the aesthetic domain, extending the orientation and calibration decomposition to genre-typed cultural contexts. Cross-institutional collaboration between the Institute for Ethics in AI (TU Munich) and KT Frontier AI Lab.

AI Technology Specialization and National Competitiveness 2024

Chun, Y., Hur, J., & Hwang, J. (2024). *PLOS ONE*. [Link](#)

The Nexus of Artificial Intelligence and Green Innovation: A Cross-Density Analysis at the Country Level 2024

Chun, Y., & Hwang, J. (2024). *Journal of the Knowledge Economy*. [Link](#)

The Moderating Role of Design Innovation in the Relationship Between Technology Complexity and Firm Performance 2024

Hur, J., **Chun, Y.**, Hwang, J., & Kim, K. (2024). *Technology Analysis & Strategic Management*, 1–14. [Link](#)

PATENTS

Method, Server and Computer Program for Dynamically Adjusting Value Information 2025 – Present

Chun, Y., Park, Y., Yoon, J., & Lee, S. (2025) [P20250217]. A value-alignment method for generative AI combining multi-agent debate and reinforcement-learning-based reward modelling to align dynamically with evolving ethical and social values without continuous human intervention.

KidnapRAG: Black-Box Attack for Hijacking Reasoning in Agentic RAG Systems Registered Jul 2026

Chun, Y. (first inventor), Choi, C., Kim, E., Lee, K., Jeong, J., Kim, E., Oh, M., Jang, J., & Chang, B. (2026). Registered patent on black-box poisoning attacks against agentic retrieval-augmented generation systems. KT-funded and filed ahead of the paper, with priority held by Chun as first inventor.

HONOURS AND AWARDS

- **Travel Grant Winner**, The Responsible AI Forum (TRAIF), Technical University of Munich — 2026.
- **Excellent Patent Award**, Invention Competition 2025, KT Corporation R&D Center — Aug 2025 (patent *Method, Server and Computer Program for Dynamically Adjusting Value Information*).
- **Best PhD Paper Award**, KOSIME 2024, Korea Society of Innovation, Management, and Economics — Jul 2024.

CONFERENCES AND SEMINARS

- **ICML 2026 Pluralistic Alignment Workshop** (Poster Session), 2026 — “Moral Orientation and Calibration: Coupled in Human Annotators, Separable in Judge LLMs.”
- **2024 Seminar on Artificial Intelligence in Budapest**, Hungary (Oct 2024) — “Strategic Approaches to AI Development and Implementation.”
- **KOSIME 2024**, Jeju, Korea (Jul 2024) — “National AI Capabilities and Product Diversification: A Multidimensional Knowledge Space Approach.”
- **6th Geography of Innovation Conference (GEOINNO 2022)**, Milan, Italy (Jul 2022) — “Economic Resilience of EU Regions to Localized Attacks.”
- **ACM SIGCHI 2020 (Fair & Responsible AI Workshop)**, Hawaii (Apr 2020) — “How Does Perceived Fairness Differ? Exploring Multicultural Landscape for Ethical AI Design.”
- **The Future Use of Autonomous Vehicles** (Invited Seminar), Paris/EHESS (Jun 2019) — “Technology Shock, Microdynamics, and Industrial Leadership Change.”

PROFESSIONAL SERVICE

Journal Reviewer

2023 – Present

Scientific Reports, PLOS ONE, Journal of the Knowledge Economy, European Research on Management and Business Economics, Business Strategy and the Environment, Research in Globalization, and Humanities and Social Sciences Communications.

AI Governance & Innovation Policy Consulting

2024 – Present

Invited Speaker, KDI (Nov 2024); AI Industry Roundtable, Seoul AI Hub (Aug 2024); AI patent/paper data analysis, KDI (Oct 2024); AI regulation research, STEPI (Jun 2024); AI-industry patent analysis, STEPI (Apr 2024).

Professional Affiliations & Networking

2024 – Present

GSMA AI for Impact Taskforce — Responsible AI Maturity Roadmap; standardised benchmarks across five key dimensions.

Policy and Industry Advisory

2023 – Present

Expert Advisor, HYGINO (carbon-footprint policy design); Advisor, National Grand Challenges Energy Division, STEPI; Advisory Committee & Author, KOTRA Global Economic and Trade Outlook Report.

TEACHING EXPERIENCE

- **Lecturer — Business Analytics**, Graduate School of Technology and Innovation Management, UNIST (Aug 2024 – Present): hands-on data analysis in Python/R, AI and generative-AI foundations, group-project supervision.
- **Invited Lecturer — Generative and Agentic AI: Innovation Mechanisms and Ethics**, Pukyong National University (Mar 2026): the evolutionary trajectory of AI from perception to generative, agentic, and physical AI; agentic AI mechanisms (multimodal interaction, reasoning/planning, action execution, self-learning); and the ethics and regulatory landscape of generative AI.
- **Invited Lecturer — AI Co-evolution and Innovation Dynamics in Mobile Technology**, Underwood International College, Yonsei University (Nov 2025): evolution of mobile/communication technologies from 1G to 5G/6G, energy and security challenges.
- **Invited Lecturer — Generative and Agentic AI: Innovation Mechanisms and Ethics**, Kyung Hee University (Mar 2025).

SKILLS

Programming: Python (ML, data science, NLP), SQL, R, LaTeX.

AI & NLP: Sentence-BERT, PatentSBERTa, RAG systems.

Analytics: network analysis, embeddings, clustering, UMAP, scientometrics, patent analytics.

Statistics & visualisation: STATA, Tableau.

LANGUAGE SKILLS

Korean — native. **English** — C1/C2 (Listening C2, Reading C2, Spoken C1, Writing C2).
Japanese — B2/C1.

REFERENCES

- **Prof. Junseok Hwang** — Professor, TEMEP, Seoul National University; Master's and Ph.D. supervisor. junhwang@snu.ac.kr
- **Prof. Keungoui Kim** — Professor, Underwood International College, Yonsei University. awekim@yonsei.ac.kr
- **Prof. Chris Russell** — Dieter Schwarz Associate Professor in AI, Government and Policy, University of Oxford (Oxford Internet Institute). chris.russell@oii.ox.ac.uk
- **Prof. Christoph Lütge** — Professor of Business Ethics; Director, Institute for Ethics in AI (IEAI), Technical University of Munich. christoph.luetge@tum.de
- **Prof. Camille Salinesi** — University Professor, Université Paris 1 Panthéon-Sorbonne; Co-Director, Observatoire de l'IA. camille.salinesi@univ-paris1.fr